

Eric Melendez

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Education

Columbia Fu Foundation School of Engineering Sep 2026 - Dec 2027
New York City, New York
MS Mechanical Engineering (Concentration in Energy Systems)

New York University Tandon School of Engineering Sep 2022 - May 2026
Brooklyn, NY GPA: 3.83
BS Mechanical Engineering; Minor Aerospace Engineering
Dean's List: F22, S23, F23, S24, F24, S25, F25, S26

Experience

Engineering Director | *NYU Rogue Aerospace* | *Brooklyn, NY* Sep 2024 - May 2026

- Oversees all engineering operations as Engineering Director for university rocketry team of 60+ students Oversees design of first ever hybrid propulsion rocket for New York University
- Designs and optimizes solid high powered rocket geometry, fin shape, recovery components, and mass distribution to target stable flight to an apogee of 4000 - 5000 ft
- Designs and manufactures parts for high powered rocketry systems using 3D printer, lasercutter, manual mill, CNC mill, and manual lathe using aluminum, steel, brass, and wood
- Validates structural designs using Ansys FEA techniques

Tech Intern | *Lavner Education* | *Austin, TX* May 2023 - Aug 2023

- Taught robotics, coding, and game design to students ages 6 - 12 in a full time capacity
- Facilitated day to day operations for a program serving 40 - 50 students
- Developed and administered instruction based on given curriculum
- Fostered interest in and understanding of science and technology in young thinkers

Projects

Payload Landing Gear | *NYU Rogue Aerospace* | *Brooklyn, NY* Sep 2024 - Apr 2025

- Worked in a small team to design a landing gear system for rocket to be used in NASA USLI
- Revised design based on prototyping to improve system reliability and ease of manufacture
- Created and maintained a working SolidWorks assembly of over fifty parts
- Created assembly and part drawings for all components for review by NASA
- Fabricated working assembly using 3D printing, laser cutting, and carbon fiber techniques

Skills

Machining Experience: Lathe (novice - intermediate), Manual Mill (novice), CNC Mill (novice)
Software: Solidworks (advanced), Fusion 360 CAM (intermediate), Ansys (intermediate), Simulink (intermediate)
Coding: Python (intermediate), Matlab (intermediate)

Licenses & Awards

Licenses: HAM Radio License (Technician)
2022 TMEA All State Jazz Band 2 (Tenor Saxophone)